FINC5001

Workshop 5

Presented by
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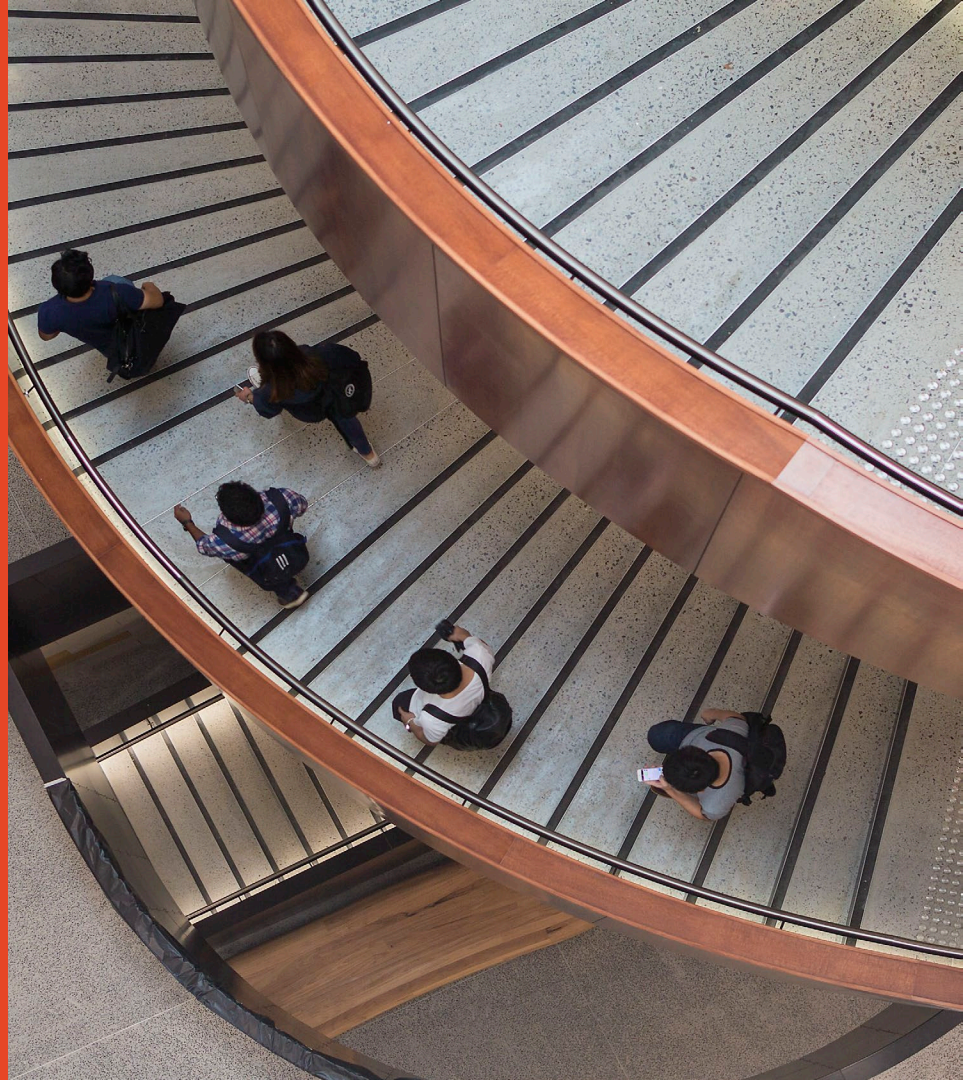

Week 6 Workshop Quiz
Can be done any time this week up until Thursday 2 April 11:59pm

Module 5 – Risk-adjusted Discounted Cash Flows, Investor Preferences, Diversification, and Portfolios

Brief Recap

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Measuring investment returns

Name of tool	Formula	What it tells you
Cash (dollar) return	$\text{Ending price, } P_{End} + \text{Cash distribution (dividend), } Div - \text{Beginning price, } P_{Beginning}$	• Measures the return from investing in a security in dollars • The higher the cash return, the greater the return earned by the investment (measured in dollars).
Rate of return, r	$\frac{\text{Cash (dollar) return}}{\text{Beginning price}} = \frac{P_{End} + Div - P_{Beginning}}{P_{Beginning}}$	• Measures the return from investing in a security as a percentage of the dollars invested • A higher rate of return means a greater return earned by the investment (measured as a percentage of the initial investment).
Expected rate of return, $E(r)$	$\left(\text{Rate of return 1, } r_1 \times \text{Probability that return 1 will occur, } \Pr(r_1)\right) + \left(\text{Rate of return 2, } r_2 \times \text{Probability that return 2 will occur, } \Pr(r_2)\right) + \cdots + \left(\text{Rate of return } n, r_n \times \text{Probability that return n will occur, } \Pr(r_n)\right)$	• The probability weighted average rate of return anticipated for an investment • The higher the expected rate of return, the greater its impact on the wealth of the investor.
Variance in the rate of return, σ^2	$\left(\text{Rate of return 1, } r_1 - \text{Expected rate of return, } E(r)\right)^2 \times \text{Probability that return 1 will occur, } \Pr(r_1) + \left(\text{Rate of return 2, } r_2 - \text{Expected rate of return, } E(r)\right)^2 \times \text{Probability that return 2 will occur, } \Pr(r_2) + \ldots + \left(\text{Rate of return } n, r_n - \text{Expected rate of return, } E(r)\right)^2 \times \text{Probability that return } n \text{ will occur, } \Pr(r_n)$	• The average variability of the rate of return • The higher the variability, in general, the greater the total riskiness of the security.

Measuring investment returns

Question being addressed:	Appropriate average calculation:
What annual rate of return can we expect for next year?	The arithmetic average rate of return calculated using annual rates of return.
What annual rate of return can we expect over a multi-year horizon?	The geometric, or compound, average rate of return calculated over a similar past period.

$$\text{Arithmetic average} = \frac{r_1 + r_2 + \ldots + r_n}{n}$$

$$\text{Geometric average return} = \left[\left(1 + \text{Rate of return for year 1, } r_{year\,1}\right) \times \left(1 + \text{Rate of return for year 2, } r_{year\,2}\right) \times \cdots \times \left(1 + \text{Rate of return for year } n\text{, } r_{year\,n}\right)\right]^{1/n} - 1$$

Present Value of Risky Cashflows

- We will incorporate risk into our analysis in two ways:

1. Expected Cashflows

$$PV = \frac{E[CF_1]}{1+r} + \frac{E[CF_2]}{(1+r)^2} + \cdots + \frac{E[CF_n]}{(1+r)^n}$$

CF_i = Cashflow in period i

r = Discount rate

$E[CF_i]$ = Expected cashflow in period i

2. Discount Rate (Required Rate of Return, Cost of Capital)

Investor Decisions Under Uncertainty

- If a risky asset has a higher expected return than a risk-free one, we call the *difference*, the risk premium of the asset. That is:

$$r_p = E(R_i) - r_f$$

r_p = Risk premium

$E(R_i)$ = Expected return on asset i

r_f = Risk–free rate

- We call this a **fair gamble**. This is where a risky investment whose expected return equals the risk-free rate of return. In other words, a risky investment that has zero risk premium.

Risk Preferences

- A **risk-averse** investor will *reject* a fair gamble. **They require an <u>appropriate</u> risk premium to accept a risky investment.** In particular, the *more* risk-averse an investor is, the *higher* the risk premium needs to be for the investor to accept the risky investment.

- A **risk-neutral** investor will be *indifferent* between a fair gamble and a risk-free alternative.

- A **risk-seeking** investor will *select* a fair gamble. They prefer a fair gamble over a risk-free alternative. Risk-seeking investors will pay a premium to take on risk.

Risk and Return

	Historical time series of data	Probability distribution of return
Expected return	$\bar{r} = \frac{\sum_{t=1}^{n} r_t}{n}$	$E(r) = \sum_{t=1}^{n} r_t \cdot p(r)$
Variance	$\sigma^2 = \frac{\sum_{t=1}^{n} (r_t - \bar{r})^2}{n-1}$	$\sigma^2 = \sum_{t=1}^{n} (r_t - \bar{r})^2 \cdot p(r)$
Covariance	$\sigma_{ij} = \frac{\sum_{t=1}^{n} (r_{i;t} - \bar{r}_i)(r_{j;t} - \bar{r}_j)}{n-1}$	$\sigma_{ij} = \sum_{t=1}^{n} (r_{i;t} - \bar{r}_i)(r_{j;t} - \bar{r}_j) p(r)$
Correlation Coefficient	$\rho = \frac{cov(i,j)}{\sigma_i \sigma_j}$	

Portfolio Mean and Variance (2-Stocks)

- The expected return of a portfolio is the **weighted average** of the expected returns of the individual stocks.

- The expected return of a portfolio with 2-stocks is:

$$ER_p = \mu_p = \omega_1\mu_1 + \omega_2\mu_2$$

- ω_1 = Portfolio weight on stock 1
- ω_2 = Portfolio weight on stock 2
- μ_1 = Expected return of stock 1
- μ_2 = Expected return of stock 2

Portfolio Mean and Variance (n-Stocks)

- The expected return of a portfolio with n-stocks is:

$$ER_p = \mu_p = \omega_1\mu_1 + \omega_2\mu_2 + \cdots + \omega_n\mu_n$$

- ω_i = Portfolio weight on stock i
- μ_i = Expected return of stock i

Portfolio Mean and Variance (2-Stocks)

- The variance of a portfolio is NOT the weighted average of the variance of the individual stocks
- The variance of a portfolio with 2-stocks is:

$$\sigma_p^2 = \omega_1^2\sigma_1^2 + \omega_2^2\sigma_2^2 + 2\omega_1\omega_2\sigma_1\sigma_2\rho_{12}$$

- σ_1^2 = Variance of stock 1
- σ_2^2 = Variance of stock 2
- σ_1 = Std dev of stock 1
- σ_2 = Std dev of stock 2
- ρ_{12} = Correlation coefficient for stocks 1 and 2

Portfolio Mean and Variance (2-Stocks)

- The standard deviation of a portfolio with 2-stocks is:

$$\sigma_p = \sqrt{\omega_1^2\sigma_1^2 + \omega_2^2\sigma_2^2 + 2\omega_1\omega_2\sigma_1\sigma_2\rho_{12}} = \sqrt{\sigma_p^2}$$

- Portfolio variance depends on the correlation (or covariance) between stocks.

- If correlation:
  - Goes up, portfolio variance goes up
  - Goes down, portfolio variance goes down

Portfolio Mean and Variance (3-Stocks)

- The variance of a portfolio with 3-stocks is:

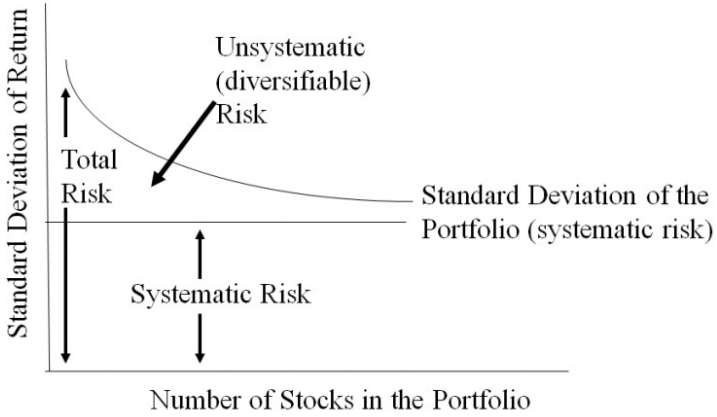
$$\sigma_p^2 = \omega_1^2\sigma_1^2 + \omega_2^2\sigma_2^2 + \omega_3^2\sigma_3^2 + 2\omega_1\omega_2\sigma_1\sigma_2\rho_{12} + 2\omega_1\omega_3\sigma_1\sigma_3\rho_{13} + 2\omega_2\omega_3\sigma_2\sigma_3\rho_{23}$$

- σ_i^2 = Variance of stock i
- σ_i = Std dev of stock i
- ρ_{ij} = Correlation coefficient for stocks i and j

Systematic and Unsystematic Risk

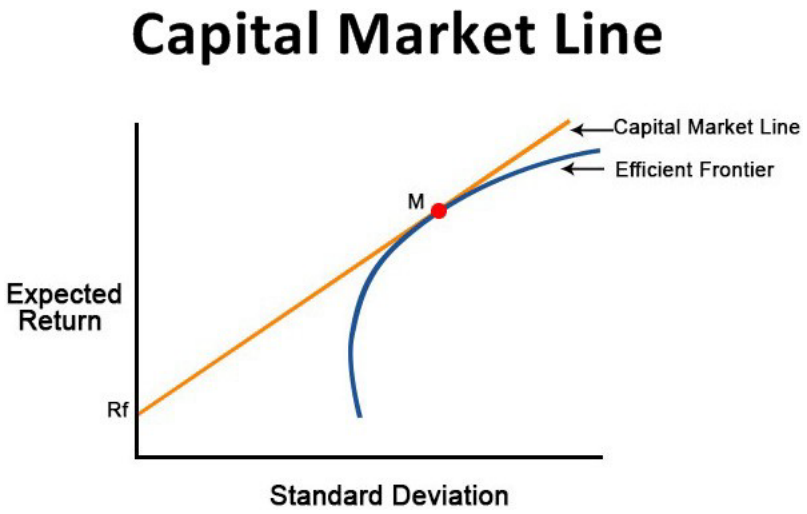
- Can you eliminate all risk? No. Diversification has its limits.

- **Systematic Risk -** Systematic risk or market risk is a risk that is common to all stocks and cannot be diversified away.
- **Unsystematic Risk -** Unsystematic risk or diversifiable risk is a risk specific to a stock that can be diversified away.



Optimal Portfolios

- **Capital Market Line (CML)** - The CML is a line that represents the risk-return tradeoff between the market portfolio and the risk-free rate.



Optimal Portfolios

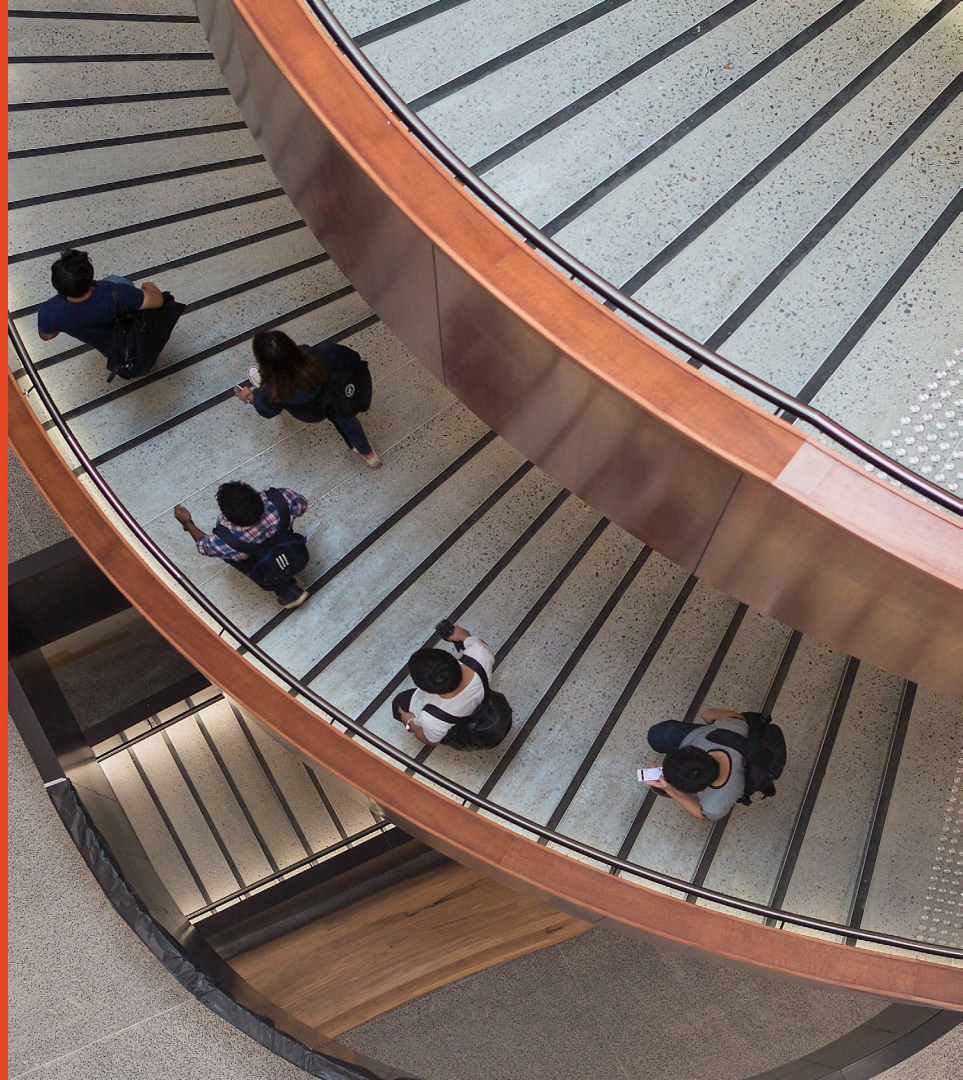
- The formula for the Sharpe Ratio is:

$$Sharpe\ Ratio_p = \frac{ER_p - r_f}{\sigma_p}$$

- ER_p = Expected return of portfolio
- r_f = Risk-free rate
- σ_p = Standard dev of portfolio

- The higher the Sharpe Ratio the better the portfolio.
- The slope of the capital market line (CML) is the Sharpe Ratio.
- The market portfolio M has the highest possible Sharpe Ratio of any portfolio on the efficient frontier.

Workshop Questions



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Question One

Explain both the covariance and the correlation between two securities. What is the difference between the two? When is one of the measures preferable to the other?

Question Two

The ordinary shares of the Boomerang Cattle Company had the following end-of-year share prices over the last five years and paid no cash dividends:

Year	Boomerang Cattle Company
1	$15
2	10
3	12
4	23
5	25

(a) *Calculate the annual rate of return for each year from this information.*

(b) *What is the arithmetic average rate of return earned by investing in the company's shares over this period?*

(c) *What is the geometric average rate of return earned by investing in the company's shares over this period?*

(d) *Which type of average rate of return best describes the average annual rate of return earned over the period (the arithmetic or geometric)? Why?*

Question Three

The following table details an analyst's prediction of the probabilities of different states of the market over the next year, along with the forecast returns on security A in each different market state. Calculate the expected return and standard deviation of returns on security A.

State	Probability	Return
Very good	0.1	**31%**
Good	0.3	**13%**
Average	0.4	**12%**
Bad	**0.2**	**-9%**

Question Four

Assets Alpha and Beta have the following estimated returns in the three possible states of the market.

	State of the Market		
	1	2	3
Return on Alpha	12%	3%	-20%
Return on Beta	4%	2%	-5%
Probability of state occurring	0.5	0.4	0.1

(a) Calculate the expected returns of assets Alpha and Beta.

(b) Calculate the variance and standard deviation of assets Alpha and Beta.

(c) Calculate the covariance and correlation coefficient between assets Alpha and Beta.

Question Five

Consider shares in two companies, X and Y, as follows:

	$E(R_i)$	σ_i
Share X	10%	20%
Share Y	30%	50%

The correlation between X and Y (i.e. r_{XY}) = 0.4

(a) What is the expected return and standard deviation of a portfolio comprising 25% invested in share X and 75% invested in share Y?

(b) Recalculate the portfolio's expected return and standard deviation assuming equal weights in X and Y.

Question Six

The following portfolios are plotted in risk-return space:

	A	B	C	D	E	F	G	H	I
E(r) (%)	4.25	5	6.25	7.75	8.25	8.75	9.5	9.5	11.25
SD (%)	12.25	11.5	10.5	12.5	14.5	14.5	16	17.5	21

a) Which of these portfolios are efficient?

b) Suppose you can also borrow and lend at a risk-free interest rate of 3.75%. Which of the above portfolios has the highest Sharpe ratio?

c) Suppose you are prepared to tolerate a standard deviation of 14.5%. What is the maximum expected return that you can achieve if you cannot borrow or lend?

d) What is your optimal strategy if you can borrow or lend at 3.75% and are prepared to tolerate a standard deviation of 14.5%? What is the maximum expected return that you can achieve with this risk?

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Covariance measures the raw degree of association between two variables. Negative covariance implies than an increase (decrease) in one variable is associated with a decrease (increase) in the other and positive covariance implies than an increase (decrease) in one variable is associated with an increase (decrease) in the other. In the case of stock returns, this is very useful in deciding which stocks we might consider combining into portfolios.

Covariance is an absolute measure and it is difficult to infer much about the relationship between the variables (other than the direction).

The correlation coefficient is a standardised variation of covariance, i.e. it is the covariance divided by the product of the standard deviations. The resulting number will always lie between +1 and –1, which provides a clear indicator of the relationship between two variables. If two stocks have a correlation coefficient of +1 then their stock returns vary uniformly. If the correlation co-efficient is -1 then stock returns completely offset one another. A more likely scenario is that stocks will be less than perfectly correlated, for example a correlation coefficient of 0.25, which implies that the stock returns will move in the same direction but not to the same extent. Most stocks have less than perfect positive correlation due to the influence of the overall market economy.

Question Two

The ordinary shares of the Boomerang Cattle Company had the following end-of-year share prices over the last five years and paid no cash dividends:

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1	$15
2	10
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4	23
5	25

(a) Calculate the annual rate of return for each year from this information.

$$\text{Rate of return} = \frac{\text{Ending price} + \text{Dividend} - \text{Beginning price}}{\text{Beginning price}}$$

$$\text{Rate of return}_{\text{Year 2}} = \frac{\$10 - \$15}{\$15} = -33.33\%$$

The remaining annual returns are summarised in the following table:

Year	Price	Return
1	$15	
2	10	-33.33%
3	12	20.00%
4	23	91.67%
5	25	8.70%

Question Two

The ordinary shares of the Boomerang Cattle Company had the following end-of-year share prices over the last five years and paid no cash dividends:

Year	Boomerang Cattle Company
1	$15
2	10
3	12
4	23
5	25

$$\text{Arithmetic average} = \frac{r_1 + r_2 + \ldots + r_n}{n}$$

$$= \frac{-0.3333 + 0.2 + 0.9167 + 0.087}{4} = 21.76\%$$

(b) What is the arithmetic average rate of return earned by investing in the company's shares over this period?

Question Two

The ordinary shares of the Boomerang Cattle Company had the following end-of-year share prices over the last five years and paid no cash dividends:

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2	10
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5	25

$$\text{Geometric average} = \left[(1+r_1)\times(1+r_2)\times\ldots\times(1+r_n)\right]^{1/n} - 1$$

$$= \begin{bmatrix} (1-0.3333)\times(1+0.20) \\ \times(1+0.9167)\times(1+0.087) \end{bmatrix}^{1/4} - 1$$

$$= 13.62\%$$

(c) What is the geometric average rate of return earned by investing in the company's shares over this period?

Question Two

The ordinary shares of the Boomerang Cattle Company had the following end-of-year share prices over the last five years and paid no cash dividends:

Year	Boomerang Cattle Company
1	$15
2	10
3	12
4	23
5	25

(d) Which type of average rate of return best describes the average annual rate of return earned over the period (the arithmetic or geometric)? Why?

This tells us that we would be indifferent between earning the actual series of Boomerang Cattle Company's returns (– 33.33%, 20%, 91.67% and 8.7%) and simply earning a constant 13.62% four times in a row. We can prove this as follows:

$$\$15(1.1362)^4 = \$25$$

If, instead, we attempted to predict our final share price by compounding the arithmetic average, we would find:

$$\$15(1.2176)^4 = \$32.97$$

Thus, if we wish to describe the multi-year behaviour of this share, we should choose the geometric average as our measure (as we always should when considering multi-year periods). However, if we wanted to estimate what we might earn from this share over the next single year (or any future single year from a comparable period), we would use our single-year estimator, the arithmetic average.

Question Three

The following table details an analyst's prediction of the probabilities of different states of the market over the next year, along with the forecast returns on security A in each different market state. Calculate the expected return and standard deviation of returns on security A.

State	Probability	Return
Very good	0.1	**31%**
Good	0.3	**13%**
Average	0.4	**12%**
Bad	**0.2**	**-9%**

Question Three

The expected return of security A given different economic state is simply the weighted average of returns with their corresponding probabilities:

$E(R_A) = (0.1 \times 31\%) + (0.3 \times 13\%) + (0.4 \times 12\%) + (0.2 \times -9\%) = 0.1$ or 10%

Variance of security A is also weighted by probabilities:

$s_A^2 = 0.1(0.31 - 0.1)^2 + 0.3(0.13 - 0.1)^2 + 0.4(0.12 - 0.1)^2 + 0.2(-0.09 - 0.1)^2$
$= 0.01206$

s_A is therefore $\sqrt{0.01206} = 0.1098$ or 10.98%

Question Four

Assets Alpha and Beta have the following estimated returns in the three possible states of the market.

	State of the Market		
	1	2	3
Return on Alpha	12%	3%	-20%
Return on Beta	4%	2%	-5%
Probability of state occurring	0.5	0.4	0.1

(a) Calculate the expected returns of assets Alpha and Beta.

(b) Calculate the variance and standard deviation of assets Alpha and Beta.

(c) Calculate the covariance and correlation coefficient between assets Alpha and Beta.

Question Four

a) $E(R_A) = (0.5 \times 12\%) + (0.4 \times 3\%) + (0.1 \times -20\%) = 5.2\%$;
$E(R_B) = (0.5 \times 4\%) + (0.4 \times 2\%) + (0.1 \times -5\%) = 2.3\%$;

b) $s_A^2 = (0.5 \times (0.12 - 0.052)^2) + (0.4 \times (0.03 - 0.052)^2) + (0.1 \times (-0.2 - 0.052)^2)$
$= 0.008856$; $\therefore s_A = \sqrt{0.008856} = 0.0941$, i.e. 9.41% p.a.

$s_B^2 = (0.5 \times (0.04 - 0.023)^2) + (0.4 \times (0.02 - 0.023)^2) + (0.1 \times (-0.05 - 0.023)^2)$
$= 0.000681$; $\therefore s_B = \sqrt{0.000681} = 0.0261$, i.e. 2.61% p.a.

c) $s_{AB} = (0.5 \times (0.12 - 0.052)*(0.04 - 0.023)) + (0.4 \times (0.03 - 0.052)*(0.02 - 0.023)) + (0.1 \times (-0.2 - 0.052)*(-0.05 - 0.023))$

$= 0.002444$

$r_{AB} = s_{AB} / (s_A \times s_B) = 0.002444 / (0.0941 \times 0.0261) = 0.995$, i.e. almost perfectly positively correlated (which makes sense looking at the expected movements in the stock returns for different market states).

Question Five

Consider shares in two companies, X and Y, as follows:

	$E(R_i)$	σ_i
Share X	10%	20%
Share Y	30%	50%

The correlation between X and Y (i.e. r_{XY}) = 0.4

(a) What is the expected return and standard deviation of a portfolio comprising 25% invested in share X and 75% invested in share Y?

(b) Recalculate the portfolio's expected return and standard deviation assuming equal weights in X and Y.

Question Five

(a) $E(R) = (0.25 \times 10\%) + (0.75 \times 30\%) = 25\%$

$s_{E(R)}^2 = ((0.25)^2 \times (0.2)^2) + ((0.75)^2 \times (0.5)^2) + (2*0.25*0.75*0.4*0.2*0.5) = 0.158125$

$\therefore\ s_{E(R)} = \sqrt{0.158125} = 0.3976$, i.e. 39.76% p.a.

(b) $E(R) = (0.5 \times 10\%) + (0.5 \times 30\%) = 20\%$

$s_{E(R)}^2 = ((0.5)^2 \times (0.2)^2) + ((0.5)^2 \times (0.5)^2) + (2*0.5*0.5*0.4*0.2*0.5) = 0.0925$

$\therefore\ s_{E(R)} = \sqrt{0.00925} = 0.3041$, i.e. 30.41% p.a.

Question Six

The following portfolios are plotted in risk-return space:

	A	B	C	D	E	F	G	H	I
E(r) (%)	4.25	5	6.25	7.75	8.25	8.75	9.5	9.5	11.25
SD (%)	12.25	11.5	10.5	12.5	14.5	14.5	16	17.5	21

a) Which of these portfolios are efficient?

b) Suppose you can also borrow and lend at a risk-free interest rate of 3.75%. Which of the above portfolios has the highest Sharpe ratio?

c) Suppose you are prepared to tolerate a standard deviation of 14.5%. What is the maximum expected return that you can achieve if you cannot borrow or lend?

d) What is your optimal strategy if you can borrow or lend at 3.75% and are prepared to tolerate a standard deviation of 14.5%? What is the maximum expected return that you can achieve with this risk?

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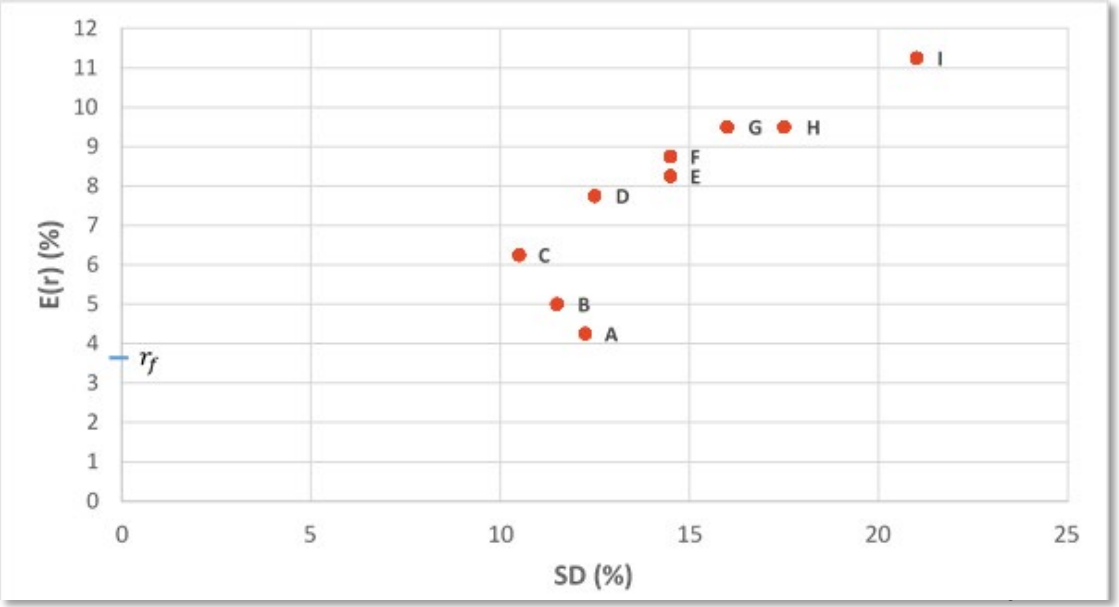
The following portfolios are plotted in risk-return space:

	A	B	C	D	E	F	G	H	I
E(r) (%)	4.25	5	6.25	7.75	8.25	8.75	9.5	9.5	11.25
SD (%)	12.25	11.5	10.5	12.5	14.5	14.5	16	17.5	21

a) Which of these portfolios are efficient?

- Portfolios A, B, E and H are **inefficient**.
- The rest are **efficient** portfolios.



Question Six *continued …*

b) Suppose you can also borrow and lend at a risk-free interest rate of 3.75%. Which of the above portfolios has the highest Sharpe ratio?

These were the efficient portfolios:

	C	D	F	G	I
E(r) (%)	6.25	7.75	8.75	9.5	11.25
SD (%)	10.5	12.5	14.5	16	21

Portfolio	$Sharpe\ ratio = \frac{ER_p - r_f}{\sigma_p}$
C	$\frac{0.0625 - 0.0375}{0.105} = 0.238095$
D	$\frac{0.0775 - 0.0375}{0.125} = 0.32$
F	$\frac{0.0875 - 0.0375}{0.145} = 0.344828$
G	$\frac{0.095 - 0.0375}{0.16} = 0.359375$ **highest**
I	$\frac{0.1125 - 0.0375}{0.21} = 0.357143$

Question Six *continued …*

c) Suppose you are prepared to tolerate a standard deviation of 14.5%. What is the maximum expected return that you can achieve if you cannot borrow or lend?

	A	B	C	D	E	F	G	H	I
E(r) (%)	4.25	5	6.25	7.75	8.25	8.75	9.5	9.5	11.25
SD (%)	12.25	11.5	10.5	12.5	14.5	14.5	16	17.5	21

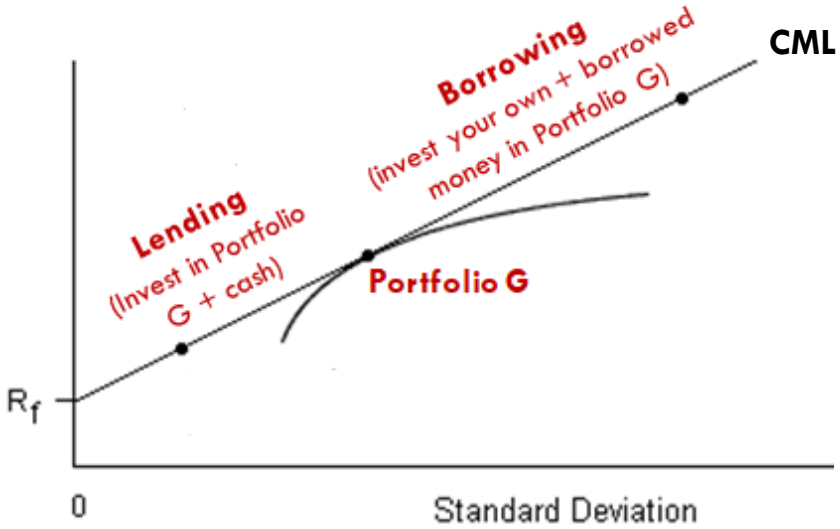
- You should select the efficient portfolio with the highest amount of risk tolerable, which is **Portfolio F.**
- This has an expected return of 8.75%.

Question Six *continued …*

	A	B	C	D	E	F	G	H	I
E(r) (%)	4.25	5	6.25	7.75	8.25	8.75	9.5	9.5	11.25
SD (%)	12.25	11.5	10.5	12.5	14.5	14.5	16	17.5	21

d) What is your optimal strategy if you can borrow or lend at 3.75% and are prepared to tolerate a standard deviation of 14.5%? What is the maximum expected return that you can achieve with this risk?

You should select some combination of **Portfolio G** and the risk-free asset.



Question Six *continued …*

d) What is your optimal strategy if you can borrow or lend at 3.75% and are prepared to tolerate a standard deviation of 14.5%? What is the maximum expected return that you can achieve with this risk?

Find the **weight** invested in portfolio G:

$$\sigma_p^2 = w_G^2\sigma_G^2 + w_{rf}^2\sigma_{rf}^2 + 2w_G w_{rf}\sigma_G\sigma_{rf}\rho_{G,rf}$$

$$(0.145)^2 = w_G^2\sigma_G^2 + 0 + 0$$

$$0.145 = w_G\sigma_G$$

$$0.145 = w_G 0.16$$

$$w_G = \frac{0.145}{0.16} = 0.90625$$

Calculate the portfolio's **expected return:**

$$ER_P = \mu_P = \omega_1\mu_1 + \omega_2\mu_2$$

$$= (0.90625 \times 0.095) + (0.09375 \times 0.0375)$$

$$= 0.08961 = 8.961\%$$

The End

Next topic we look at the Capital Asset Price Model (CAPM)